

Executive Summary

The analysis of this report provides indicative option prices and warning about the direction of future interest rates. The independent pricing checks give similar results confirming the overall reliability of the pricing mechanism even though some internal estimates may vary.

The key findings were as follows;

For a 20-day Asian Call, the fair value was \$4.83 and a 4% bank charge of % 0.19 bringing the total price to \$5.02. At 95% confidence level, the simulated prices fall between \$4.79 and \$4.86

For the 70-day Put, the fair value was \$9.54 with a 4% bank charge of 0.38% bringing the client price to \$9.92. At 95% confidence level, the simulated prices fall between \$9.46 and \$9.61.

The current market prices were used to first understand the short-term environment and the results were then used to estimate prices for the request client contracts. Lastly, the interest rates analysis was added as a forward looking approaching to ascertain the future sources of risk and how such could affect future funding and cash flows. Below is the diagrammatic illustration of the project workflow

SM Share Option Pricing Mechanism: End-to-End Workflow

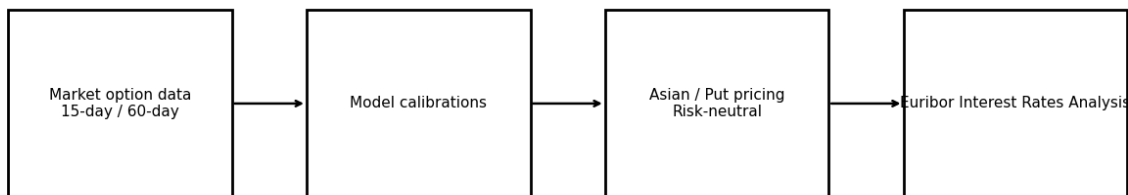


Figure 1: Project workflow

1 (a) 15-day Maturity

The graph shows observed market prices and their corresponding calculated or model price for different strike prices. Generally, the model prices follow the market dynamics which means that the valuation can be adopted as an indicative benchmark.

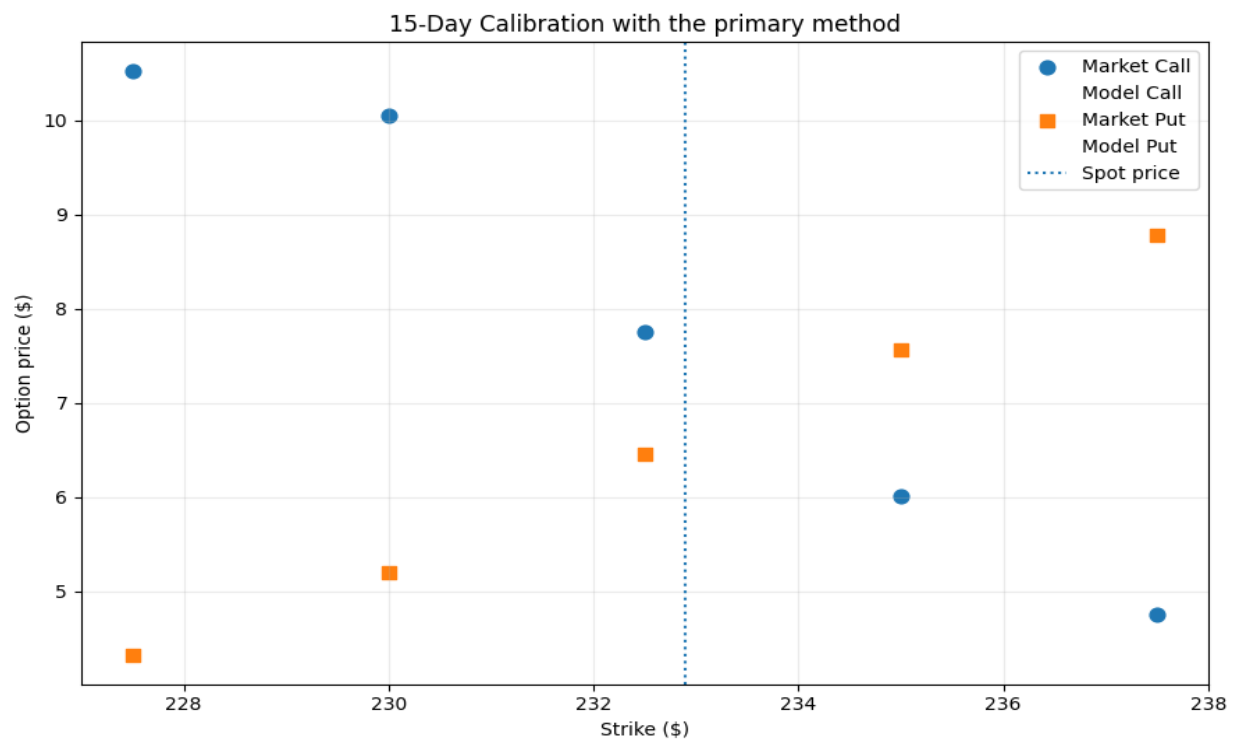


Figure 2: Observed and fitted 15-day option prices

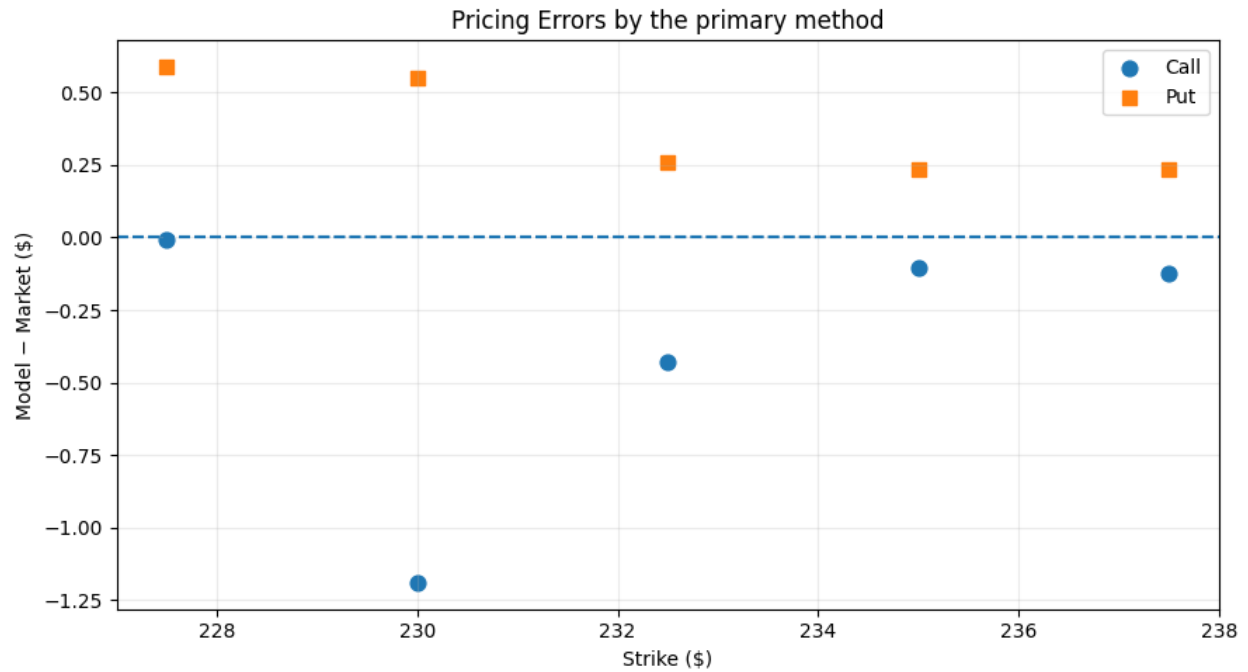


Figure 3: Pricing differences for the 15-day options

The differences between the calculated prices and the quoted market process are close to zero. This is useful to show management where the valuation is less accurate.

Measure	Primary calculation	Alternative calculation
Mean squared pricing error	0.246154	0.247459
Average error in price terms	\$0.496	\$0.497
Largest difference between calculated prices	—	\$0.140

1 (b) Primary Calculation Vs Alternative Calculation

The estimates produced by the primary and the alternative calculations were compared and the starting variance and mean-reversion speed were very close while volatility and correlation returned slightly larger differences. The two methods can produce similar prices even when some underlying parameters are not necessarily identical. This underscores the fact that the consistency of the resulting option values is more important than treating each individual parameter estimate as exact

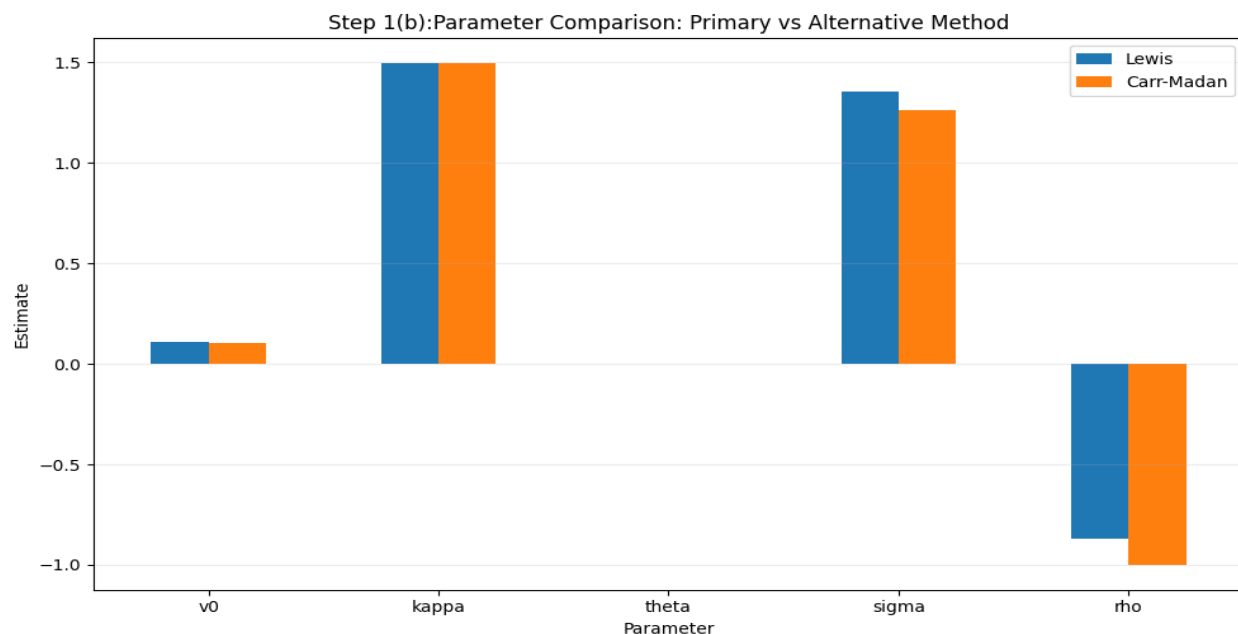


Figure 4: Parameter comparison

1 (c) 20-day Asian call

The value of Asian options is path dependent and it is obtained by the average of the share price during the contract period.

Item	Result
Current share price	\$232.90
Contract type	20-day at-the-money Asian call
Estimated fair value	\$4.83
Simulation standard error	\$0.019
95% simulation range	\$4.79 – \$4.86
4% bank fee	\$0.19
Indicative client price	\$5.02

The option fair value is the economic value of the contract before the bank charges. For transparency, the bank charge of 4% has been shown separately such that if added to the fair market value, a final price can be arrived at. These results were obtained after a large number of simulations which means that, the results are fairly stable.

A distribution of possible financial outcomes for the contract is shown below;

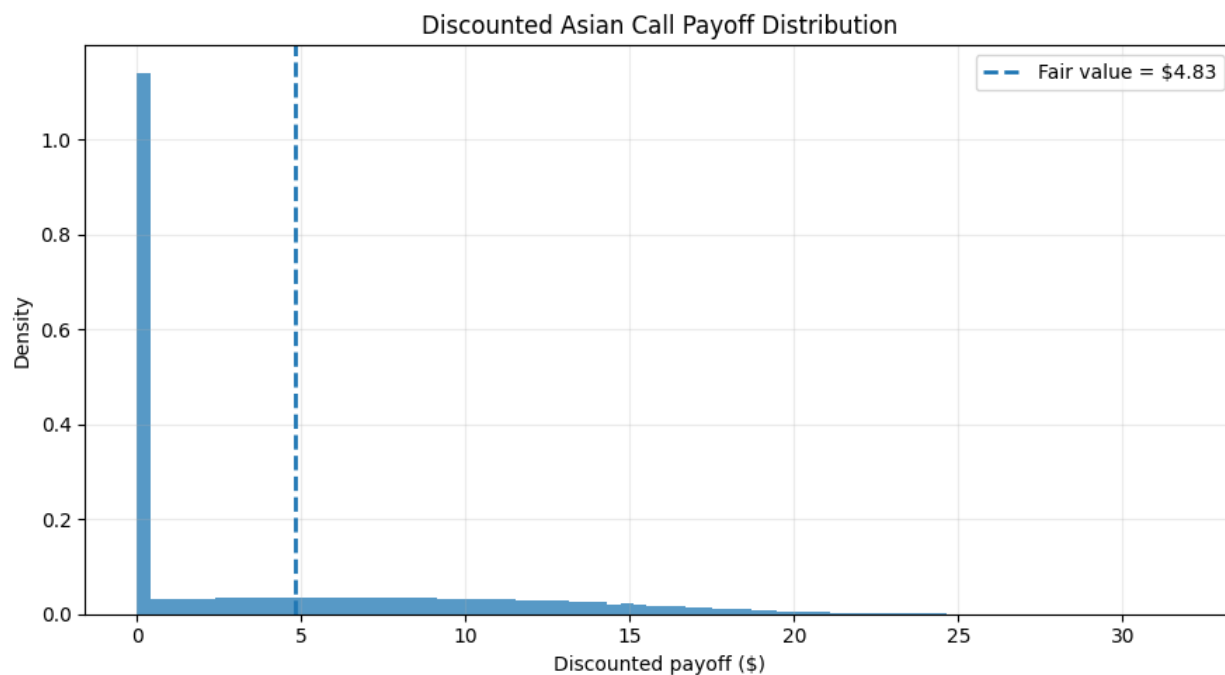


Figure 5: 60-Day Asian call payoff Distribution

The dashed line represents the central estimate which is an average of the discounted payoffs across simulated paths. In this case it was 4.83 which is highly supported by a very small simulation uncertainty.

Step 2 (a) 60-Day Calibration with the primary method

Longer periods usually expose share prices to large unusual and sudden jumps or movements. The framework was therefore expanded to allow for both ordinary or normal variations and sudden price swings.

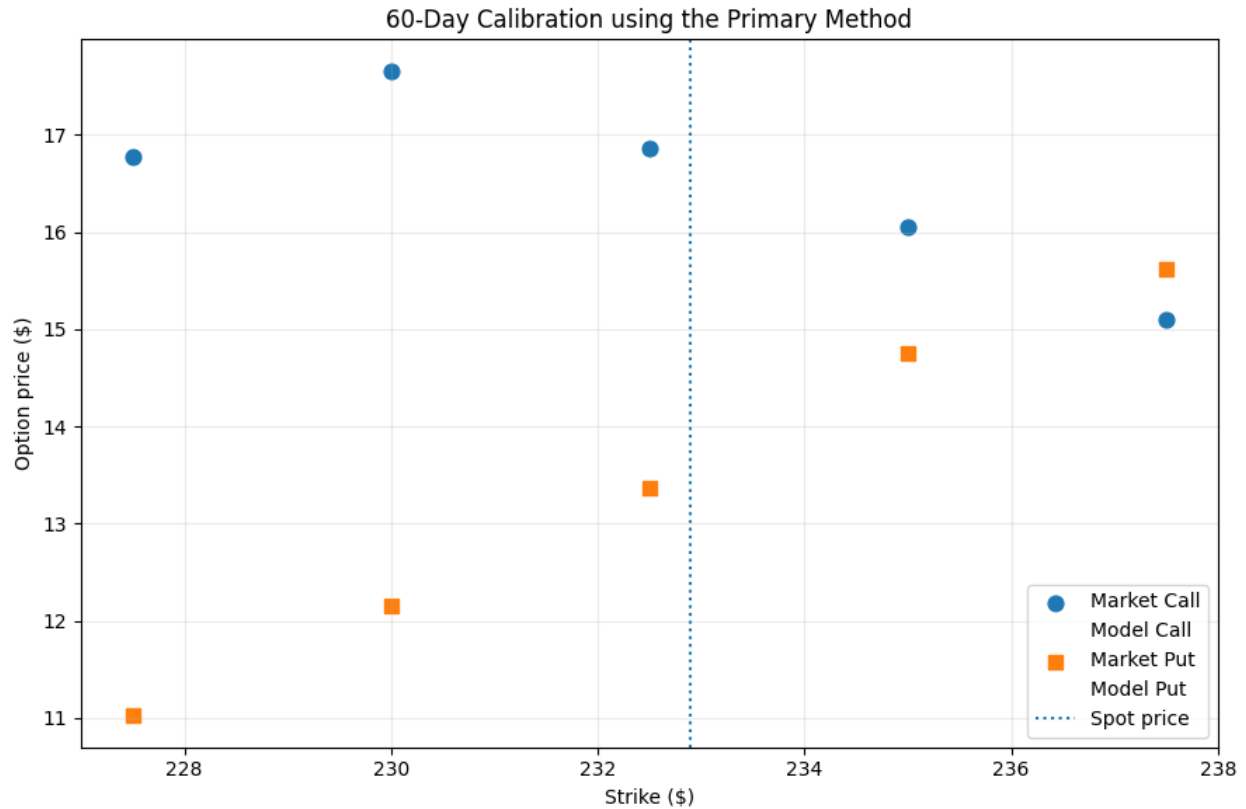


Figure 6: 60-Day calibration with the primary method

Overall, the model values follow the market pattern, but model's ability to match market values declines as maturity increases.

2 (b) 60-Day Calibration with the alternative method

Measure	60-day result
Average pricing error in price terms	\$1.150
Mean squared pricing error	1.323
Alternative calculation MSE	1.335
Largest Lewis/alternative price difference	\$0.054

The results in the table above are a signal to the bank that the 60-day market is more difficult to fit using the available observations as shown by its higher pricing error compared to the 15-day framework. At the same time, the alternative pricing method gives an overall error that is close to that of the primary method. This means that even though longer maturities carry more model and market risk, the valuation is still reasonably robust.

2 (c) 70-Day Calibration

The client requested for a 70-day Put with moneyness of 95%, therefore the strike price was set at \$221.25 since the current share price of SM is \$232.90

Item	Result
Current share price	\$232.90
Strike	\$221.25
Maturity	70 days
Estimated fair value	\$9.54
Simulation standard error	\$0.039
95% simulation range	\$9.46 – \$9.61
4% bank fee	\$0.38
Indicative client price	\$9.92

The estimated fair value of the option is \$9.54 but the client would have to pay approximately \$9.92 after adding a 4% bank charge.

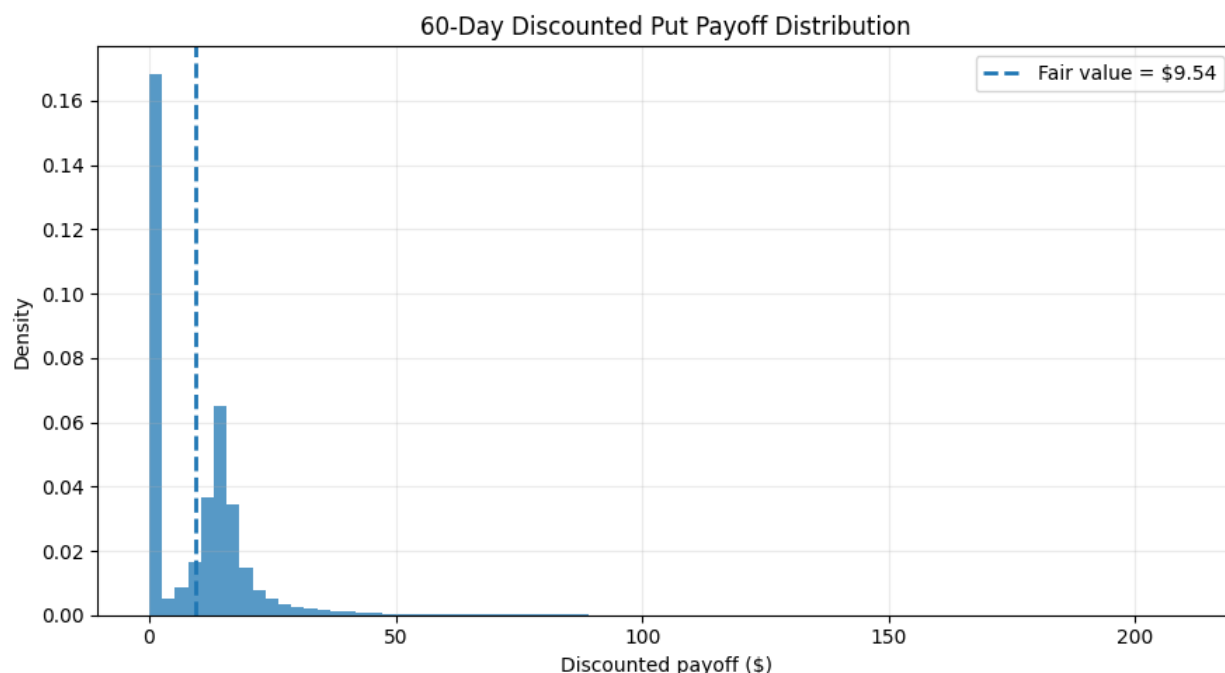


Figure 7: Discounted Put Payoff Distribution

The distribution shows the range of possible values created by different future share price outcomes. It also shows downside sensitivity in that if the share price falls materially, the bank's exposure increases. It is therefore crucial to monitor downside risk after booking the trade. Therefore, the position should not only be assessed solely on its fair value, the exposure too has to be assessed and monitored because it can materially increase as share prices fall greatly.

Step 3: Interest-Rate Outlook

This section examines how future interest rates could affect pricing and funding. The following current Euribor rates and maturities were used to form a smooth one-year curve which was then used to generate future scenarios.

Maturity	Current Euribor
1 week	0.648%
1 month	0.679%
3 months	1.173%
6 months	1.809%
12 months	2.556%

The current Euribor term structure is illustrated by the graph below;

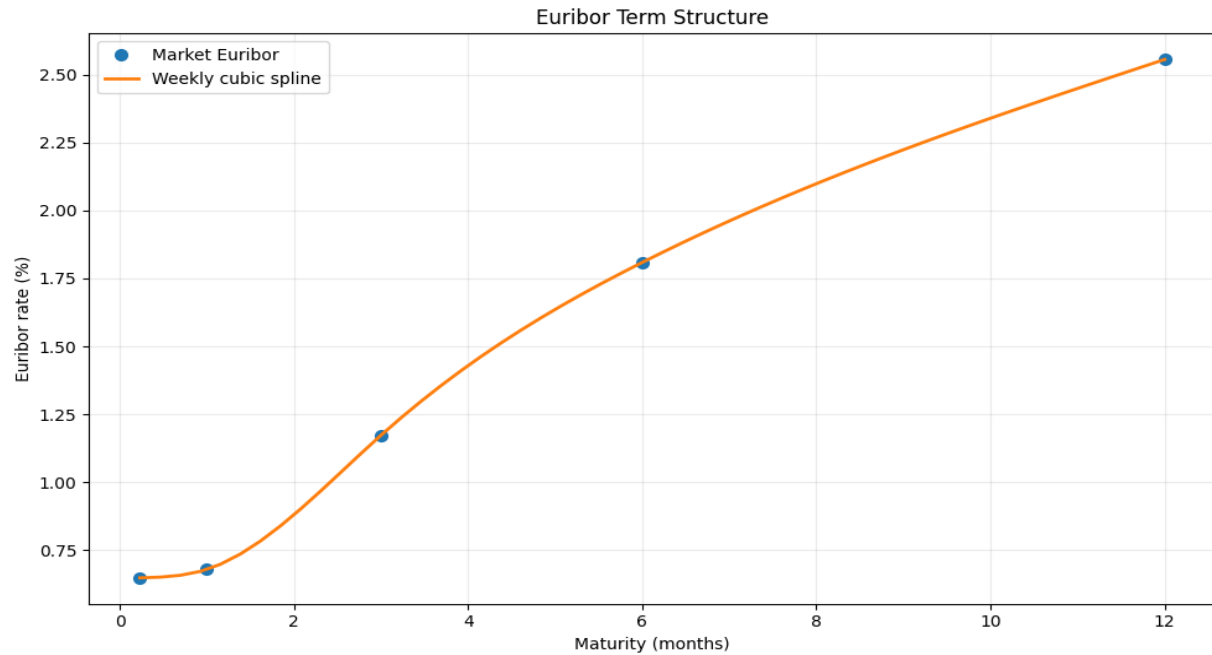


Figure 8. Current Euribor term structure and weekly interpolation.

The market observations are connected by a smooth line with a goal of trying to avoid treating the quoted maturities as isolated points. This provides a consistent picture of today's interest rate environment for scenario analysis.

Graphical representation of the Interpolated Euribor Curve

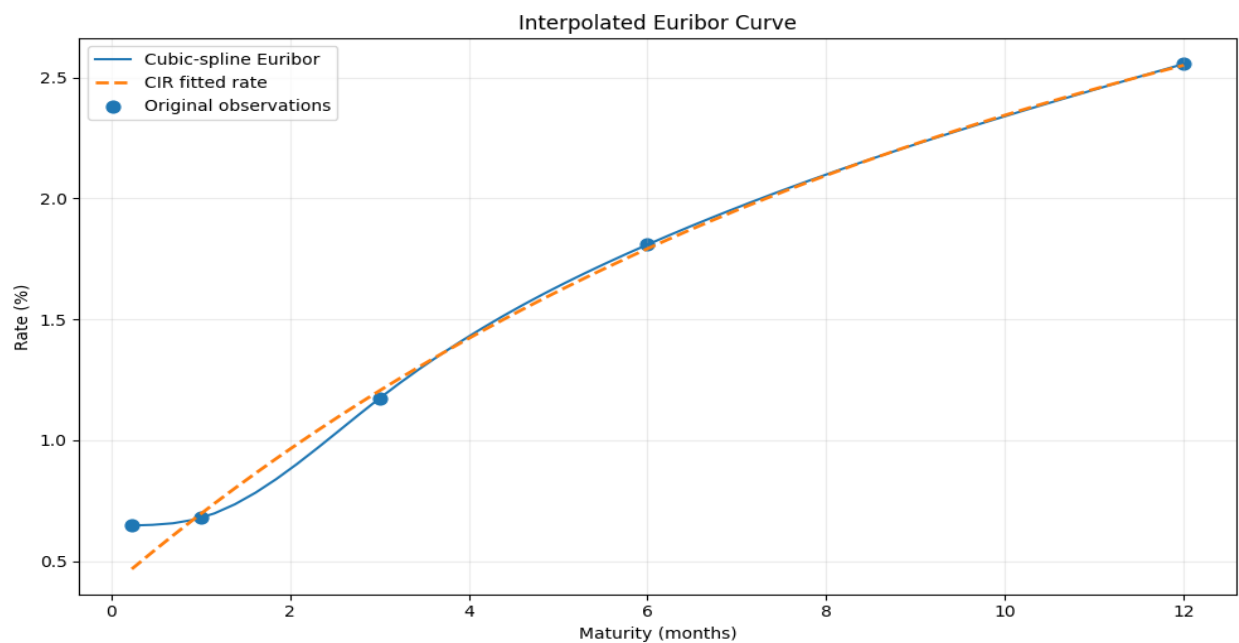


Figure 9. Fitted interest-rate curve.

The fitted line follows the overall shape of the market curve reasonably well, therefore serves as the practical starting point for future rates.

3 (b) One-Year Interest-Rate Scenarios

A total of one hundred thousand thousand daily scenarios were generated for one year and this exercise produces a range of possible future 12-month Euribor rates. A 95% confidence is used as the management scenario range

Measure	Result
Expected 12-month Euribor in one year	3.68%
95% lower bound	2.70%
95% upper bound	4.82%
Current 12-month Euribor	2.556%
Expected increase	112.6 basis points

The expected one-year rate is above today's 12-month Euribor by about 113 basis points. The 95% range is wide enough to show that the bank should not plan around one exact future rate. Instead, pricing and funding decisions should be tested against both the central expectation and the adverse higher-rate scenarios.

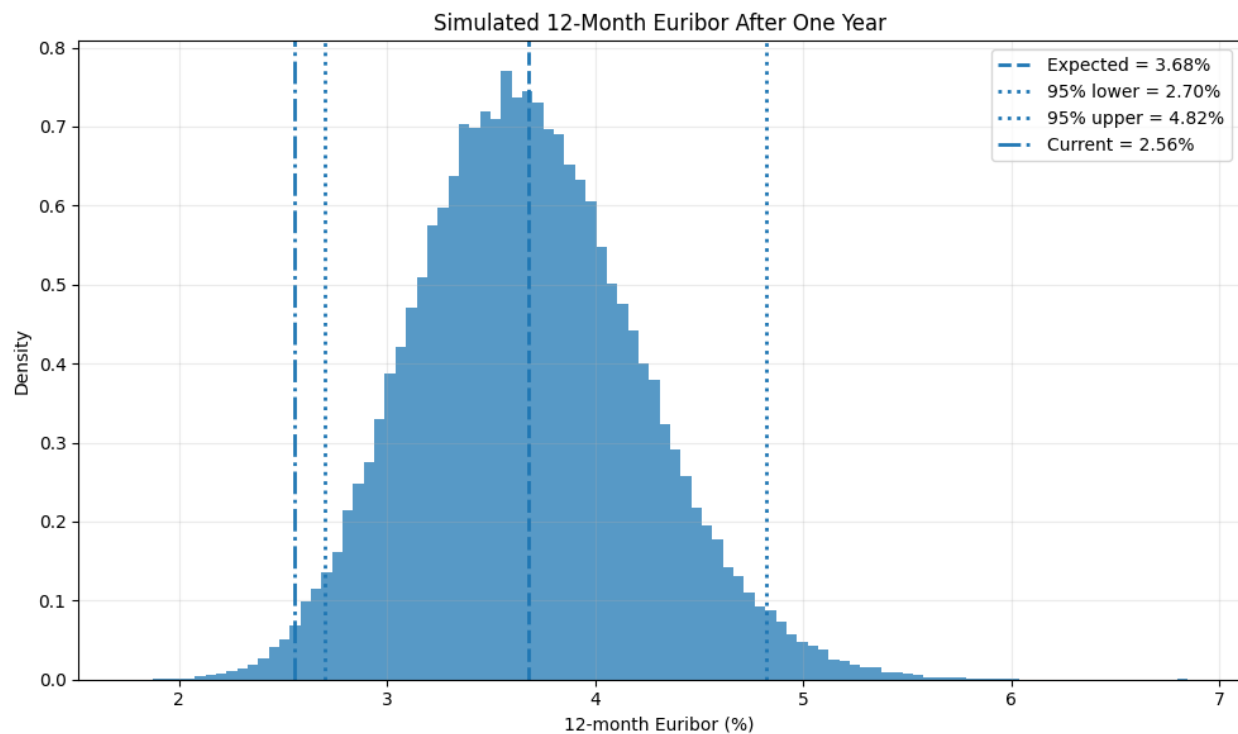


Figure 10. One-year-ahead distribution of 12-month Euribor.

The chart shows a wide range of possible future rates generated by scenario analysis. The central value is the expected rate while the two outer markers identify the selected 95% range. This means that higher rates are a meaningful possibility and not a far-fetched outcome and therefore deserve consideration in future pricing and funding plans.

Today's 12-month Euribor is 2.556%, while the expected one-year-ahead level is 3.68%. If this scenario materializes, products that depend on future borrowing, funding or discounting would face a higher interest-rate environment. The exact effect depends on each product's cash flows: some positions are hurt by higher rates, while others may benefit. The bank should therefore use the scenario as an input to product-specific pricing rather than applying one blanket adjustment.

Conclusion

The analysis gives indicate client prices of \$5.02 for the 20-day Asian call and \$9.92 for the 70-day Asian Put, these prices include a 4% bank charge. The pricing checks are reasonably consistent although longer maturities carry greater valuation uncertainty.

The interest rates analysis indicates a possible rise in the 12-month Euribor from 2.56% to 3.68% which could increase future funding and pricing costs. The bank should therefore update market conditions before final pricing and maintain appropriate risk controls

References

Bates, David S. "Jumps and Stochastic Volatility: Exchange Rate Processes Implicit in Deutsche Mark Options." *The Review of Financial Studies*, vol. 9, no. 1, 1996, pp. 69–107.

Carr, Peter, and Dilip B. Madan. "Option Valuation Using the Fast Fourier Transform." *The Journal of Computational Finance*, vol. 2, no. 4, 1999, pp. 61–73.

Cox, John C., Jonathan E. Ingersoll, Jr., and Stephen A. Ross. "A Theory of the Term Structure of Interest Rates." *Econometrica*, vol. 53, no. 2, 1985, pp. 385–407.

Heston, Steven L. "A Closed-Form Solution for Options with Stochastic Volatility with Applications to Bond and Currency Options." *The Review of Financial Studies*, vol. 6, no. 2, 1993, pp. 327–343.

Hull, John C. *Options, Futures, and Other Derivatives*. 11th ed., Pearson, 2022.

Lewis, Alan L. *A Simple Option Formula for General Jump-Diffusion and Other Exponential Lévy Processes*. 2001.

Rubinstein, Mark. "Implied Binomial Trees." *The Journal of Finance*, vol. 49, no. 3, 1994, pp. 771–818.